www.tntextbooks.in

In cortical nephrons, blood from efferent arteriole flows into peritubular capillary beds and enters the venous system carrying with it recovered solutes and water from the interstitial fluid that surrounds the tubule.

6MNMLS

ii) Tubular Reabsorption

This involves movement of the filtrate back into the circulation. The volume of filtrate formed per day is around 170-180 L and the urine released is around 1.5 L per day, i.e., nearly 99% of the glomerular filtrate that has to be reabsorbed by the renal tubules as it contains certain substances needed by the body. This process is called selective reabsorption. Reabsorption takes place by the tubular epithelial cells in different segments of the nephron either by active transport or passive transport, diffusion and osmosis.

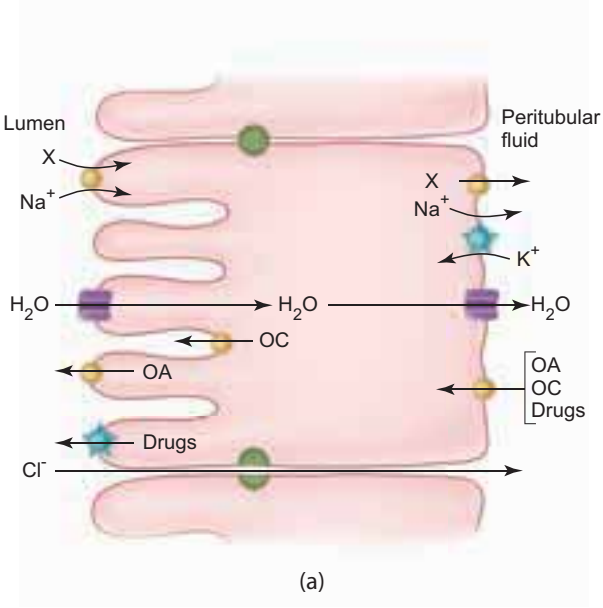

Figure 8.9 (a) Transport in the proximal convoluted tubule cells
(OA: Organic anion. OC: Organic cation).

Proximal convoluted Tubule (PCT)- Glucose, lactate, amino acids, Na^1 and water in the filtrate is reabsorbed in the PCT. Sodium is reabsorbed by active transport through sodium- potassium (Na^1 K^1) pump in the PCT. Small amounts of urea and uric acid are also reabsorbed.

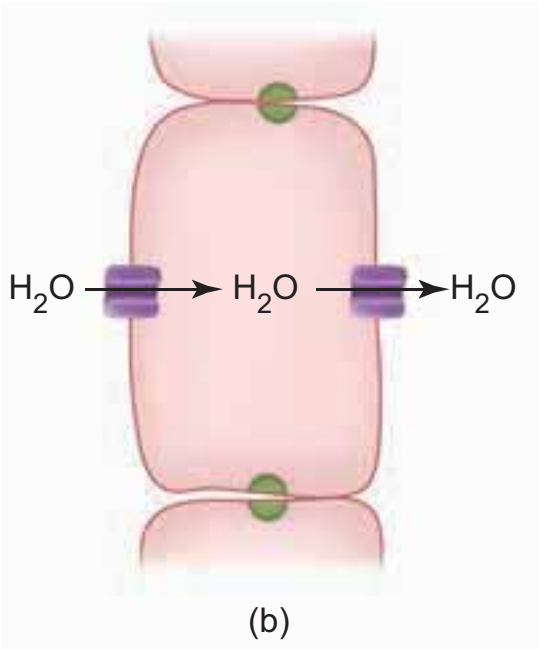

Figure 8.9 (b) Thin descending limb

Figure 8.9 (a) shows the transport in the proximal convoluted tubule cells.

Descending limb of Henle's loop is permeable to water due the presence of aquaporins, but not permeable to salts. Water is lost in the descending limb, hence Na^1 and Cl^2 gets concentrated in the filtrate (Figure 8.9(b)).

Ascending limb of Henle's loop is impermeable to water but permeable to solutes such as Na^1, Cl^2 and K^1 (Figure 8.9(c)).

The **distal convoluted tubule** recovers water and secretes potassium into the tubule. Na^1, Cl^2 and water remains in